

Conditional Edges in Trend-Filtered Mean-Reversion: A Six-Condition Redesign

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Abstract

This report documents the "Six-Condition Redesign": a pre-registered, three-stage statistical gauntlet built to test whether **trend-filtered mean-reversion** — a rule-based strategy combining a 252-day trend filter with a Bollinger-style mean-reversion trigger and a full risk overlay (stop-loss/take-profit, volatility-based sizing, a drawdown circuit breaker) — has an edge that depends on the prevailing market regime, even though the rule's own *unconditional* edge did not survive this project's standard two-part significance test. Six market-state conditions are constructed for every traded asset on a strictly point-in-time, no-lookahead basis: trailing volatility regime, trend strength, credit-spread regime, interest-rate environment, structural breaks (sparse SADF), and S&P 500 market breadth (percentage of constituents above their own 200-day moving average, built from a fresh CRSP/WRDS pull). Each asset-condition-group cell is tested three ways: a price-path-permutation Monte Carlo test (MCPT), a pooled Deflated Sharpe Ratio (DSR) correction across every cell tested, and — for the higher-order hypothesis that a *joint*, nonlinear combination of all six conditions might predict individual trade outcomes even where no single condition does — a purged, embargoed cross-validated Random Forest meta-labeling classifier.

The gauntlet was run in full, with 100% real data, on two structurally different universes (14 cross-asset ETFs; 10 sector ETFs). The result is a decisive, three-way-confirmed null: no cell in either universe clears both the permutation test and the pooled DSR correction (best case, 14-asset universe: PSR = 0.630 against a 0.95 bar), and the meta-labeling classifier loses to a naive base-rate baseline in essentially every cross-validated fold in both universes. One descriptive wrinkle — market breadth's unusually large feature importance in the 14-asset universe — does not replicate in the 10-sector universe and does not change either universe's pass/fail outcome, and is reported as a footnote rather than a finding. Per a standing project directive, this outcome is reported as the final answer rather than used as a trigger to search for a better-looking feature, model, or strategy.

Keywords: trend following, mean reversion, Monte Carlo permutation test, deflated Sharpe ratio, meta-labeling, purged cross-validation, market regime conditioning, multiple-testing correction

1. Introduction

Most single-factor trading rules are tested unconditionally: does the rule make money, on average, across all market conditions? A rule can fail that test and still contain a genuine, narrower edge — one that only appears in a specific market regime and is diluted to insignificance when averaged across all regimes. This report asks the narrower question directly: **conditional on the prevailing market state, does trend-filtered mean-reversion have an edge that a rigorous, pre-registered testing procedure can detect?**

The rule under test, trend-filtered mean-reversion, was originally designed with risk management built in from the start rather than retrofitted afterward: a 252-day trend filter restricts mean-reversion entries to the direction of the asset's own longer-term trend, combined with a standard 20-day/2-standard-deviation Bollinger-style entry trigger, a stop-loss/take-profit exit, volatility-based position sizing, and a portfolio-level drawdown circuit breaker. Tested unconditionally on the 14-asset universe described below, the rule showed a genuinely encouraging permutation-test result (5 of 14 assets significant at $p < 0.05$ on Sharpe ratio) that did **not** survive the mandatory Deflated Sharpe Ratio correction (0 of 14 assets at the 0.95 threshold) — a result that is itself methodologically instructive, since a naive stop at the permutation-test stage alone would have wrongly suggested a real discovery.

Rather than treating that unconditional null as the end of the inquiry, this redesign asks whether the same rule's edge is regime-dependent: perhaps trend-filtered mean-reversion works well in low-volatility, strongly-trending, tight-credit-spread environments and poorly outside them, in a way that washes out unconditionally but would show up once trades are split by market state. This is a **pre-registered** design: the six conditioning variables, the three testing stages, and the pass/fail bar for each were fixed before any real-data results were seen, and the redesign was carried out under an explicit standing methodological discipline — **no changing strategies because a result comes back null, and no changing methods (MCPT, DSR, purged cross-validation) because they are not producing a promising-looking result**. This report is the record of that discipline being followed to its conclusion, including where the conclusion is a decisive null.

2. Strategy Specification and Trading Universes

Trading rule. Trend-filtered mean-reversion combines two conditions on each asset's own price history: (i) a 252-day trend filter, which restricts eligible trades to the direction of the asset's own trailing 252-day trend, and (ii) a standard 20-day, ± 2 -standard-deviation Bollinger-style mean-reversion trigger. A full risk-management overlay is applied on top: a stop-loss and take-profit exit, position sizing scaled to trailing realized volatility, and a

portfolio-level drawdown circuit breaker (sized off a "shadow" hypothetical uninterrupted equity path rather than the breaker's own realized, already-cut equity, to avoid the breaker latching permanently on after a single trigger).

Bet construction. Every completed round-trip trade — from entry (t_0) to exit (t_1), whether by take-profit, stop-loss, or the rule's own timing — is one observation ("bet"). Each bet carries `trade_pnl` (realized profit or loss) and a binary `label` (1 if `trade_pnl > 0`, else 0), which doubles as the meta-labeling target in Stage 3 without any further triple-barrier construction, since the rule's own exit logic already defines the barrier.

Universes. The redesign is run on two structurally different universes, so that any conditional edge found on one cannot be dismissed as an artifact of that universe's own composition:

- **14-asset cross-asset universe:** SPY, QQQ, IWM, EFA, EEM (equities); TLT, IEF, SHY, HYG, LQD (fixed income); GLD, SLV, DBC (commodities); VNQ (real assets) — 1,309 completed trades over the full tradeable history common to this panel.
- **10-sector-ETF universe:** the S&P sector SPDRs (XLE, XLF, XLK, XLV, XLI, XLP, XLY, XLU, XLB, XLRE) — 1,051 completed trades.

The two universes maintain separate testing pools throughout (separate DSR corrections, separate meta-labeling runs), since the Deflated Sharpe Ratio corrects for multiple tests conducted *within* one universe's own search, not across two independently designed studies.

3. The Six Conditioning Variables

Every condition is constructed to the same standard: a **per-asset, point-in-time expanding median split** into HIGH (above the asset's own trailing median) or LOW (at or below it), computed only from each asset's own strictly-prior history (today's value is never included in today's own median), and gated by a 252-calendar-day-since-inception warm-up period during which no label is assigned, to avoid conditioning on an unstable small-sample median early in an asset's history.

Condition	What it measures	Source	Update frequency
<code>vol_regime</code>	Trailing realized volatility regime	The tested asset's own return history	Daily
<code>trend_strength</code>	Magnitude of the 252-day trend filter's own signal	The tested asset's own return history	Daily

Condition	What it measures	Source	Update frequency
<code>credit_spread</code>	Credit-market stress regime	HYG/LQD (external, market-wide — never treated as "the tested universe" even when HYG/LQD are themselves traded assets)	Daily
<code>rate_environment</code>	Interest-rate level/curve regime	US Treasury yields, FRED (DGS10, DGS3MO)	Daily
<code>structural_break</code>	Explosiveness / regime-change detection	Sparse SADF (supremum Augmented Dickey-Fuller) test, evaluated only at sparse test dates, not daily	Sparse
<code>market_breadth</code>	Breadth of the broad equity market	<code>pct_above_200dma</code> : the fraction of point-in-time S&P 500 constituents trading above their own 200-day moving average	Daily

Market breadth construction (the newly added sixth condition). Built from a fresh CRSP/WRDS pull of point-in-time S&P 500 membership and constituent daily prices, split-adjusted (`dlyprc / dlycumfacpr`, the same correction already established for this project's options-overlay work). A constituent counts on a given date only if it was a genuine index member that day (no survivorship bias) *and* its own 200-day moving average is already defined (at least 200 valid prior trading days of its own history) — mirroring this project's staggered-inception principle elsewhere. Three real CRSP data-quality patterns were found and handled explicitly on the real pull, none of them treated as reasons to drop or interpolate an observation: a `NaN` price (no quote that day) is left as missing; a negative price (CRSP's own bid-ask-average, no-trade convention) is recovered via absolute value; an invalid (`NaN` or non-positive) adjustment factor produces a missing, not a fabricated, adjusted price.

Sample composition, both universes (completed trades by condition, HIGH / LOW / missing):

Condition	14-asset HIGH	14-asset LOW	14-asset NaN	10-sector HIGH	10-sector LOW	10-sector NaN
<code>vol_regime</code>	629	635	45	505	503	43
<code>trend_strength</code>	431	833	45	440	568	43

Condition	14-asset HIGH	14-asset LOW	14-asset NaN	10-sector HIGH	10-sector LOW	10-sector NaN
<code>credit_spread</code>	528	515	266	355	379	317
<code>rate_environment</code>	397	899	13	409	637	5
<code>structural_break</code>	613	682	14	536	505	10
<code>market_breadth</code>	619	657	33	531	478	42

(14-asset total: 1,309 trades; 10-sector total: 1,051 trades.) `credit_spread` 's larger missing count is mechanical, not a data problem: HYG (2007) and LQD (2002) both post-date several other assets' own inception, so credit spread is genuinely undefined for trades before HYG/LQD existed, independent of the traded asset's own warm-up.

`trend_strength` 's skew toward LOW is an expected property of a point-in-time (not full-sample) expanding median on a non-stationary series: a trending regime can push an asset's own trend strength above its still-lagging running median for extended stretches.

4. Methodology: The Three-Stage Significance Gauntlet

Three independent, pre-registered stages, run in sequence, each asking a genuinely different question:

Stage 1 — Price-path-permutation Monte Carlo test (MCPT). For each (asset, condition, group) cell, is the group's realized Sharpe ratio and cumulative return real, or is it what a random price path with the same statistical properties would produce anyway? Each trial reconstructs a synthetic price path from the asset's own shuffled returns and **reruns the entire signal → trade → condition-labeling chain from scratch** — not a simpler label-permutation shortcut — because three of the six conditions (`vol_regime` , `trend_strength` , `structural_break`) are themselves computed from the tested asset's own price history and must be recomputed consistently with each permuted path. Two conditions — the external yield series behind `rate_environment` / `credit_spread` , and the newly added `market_breadth` — are genuinely external, market-wide signals with no dependence on any one tested asset's own price, and are held constant across every trial rather than permuted.

The number of trials (`NREPS`) is fixed at 30 rather than this project's standing convention of 2,000, because `structural_break` 's SADF evaluation dominates per-trial cost, making 2,000 trials infeasible within this project's compute budget. This number was chosen deliberately, not casually: an earlier version of this module picked `NREPS` live from a runtime benchmark, which produced two different trial counts (37 and 34) on two different machines running the identical test with the identical seed — and therefore two different sets of nominally significant cells, sharing only 8 of 11–17 flagged cells between them. This is a genuine reproducibility defect (a Monte Carlo test's trial count should

never depend on whose machine ran it), caught by directly diffing the two runs rather than trusting either one, and fixed by making $NREPS = 30$ a fixed constant, reproducible on any machine with the same seed (20260930) going forward.

Each universe's full test surface: 14 assets $\times$ 6 conditions $\times$ 2 groups = **168 tests** (14-asset universe); 10 assets $\times$ 6 conditions $\times$ 2 groups = **120 tests** (10-sector universe) — a large multiple-testing surface, addressed explicitly by Stage 2.

Stage 2 — Deflated Sharpe Ratio (DSR) pooled correction. Reuses the Bailey & López de Prado (2014) extreme-value-theory framework, unchanged from its earlier use elsewhere in this project: an expected-maximum-Sharpe benchmark (SR_0) is computed once across **all** trial Sharpe ratios in a universe's own pool (168 or 120), and each individual trial's Probabilistic Sharpe Ratio (PSR) — using that trial's own sample length, skewness, and kurtosis — is tested against SR_0 at a 0.95 threshold. **An edge is accepted as real only if it clears both bars:** nominal MCPT significance ($p < 0.05$) and $PSR \geq 0.95$ against the pooled SR_0 — this project's standing two-part rule, applied here exactly as it was applied earlier to the same rule's own unconditional test. A flagged, honest limitation: many of the trials pooled into one SR_0 are not fully independent draws by construction — HIGH and LOW within one condition are complementary partitions of the same asset's own trades, and the same asset recurs across all six conditions — which means the nominal trial count somewhat overstates the number of genuinely distinct hypotheses tested, even though SR_0 itself still reflects the trials' actual empirical dispersion.

Stage 3 — Percentile-rank meta-labeling. Even if no single condition shows a univariate edge, a *joint*, nonlinear combination of all six might still predict which of the rule's own trades will be profitable — a genuinely different hypothesis from Stages 1–2, not a re-test of the same one. The feature representation is each condition's own **expanding percentile rank** at the trade's decision date:

$$\text{rank}_t = \frac{\#\{s < t : x_s \leq x_t\}}{\#\{s < t\}}$$

the fraction of an asset's own strictly-prior valid history that is less than or equal to today's value (0.5 sits exactly at the Stage 1–2 median-split boundary). This was chosen over two alternatives considered and rejected: raw absolute condition values, which would reintroduce the cross-asset, cross-era scale-pooling problem the per-asset-relative median-split design was built to avoid; and the existing ± 1 HIGH/LOW label, which discards magnitude and distance-from-median information a joint classifier could otherwise exploit. Ranks are evaluated only at trade decision dates, not daily, since meta-labeling needs one feature value per trade, not a continuous daily series.

The model and cross-validation scheme are reused **verbatim** from this project's own already-established meta-labeling architecture, deliberately, per the standing no-method-shopping discipline: combinatorial purged K-fold cross-validation ($N_SPLITS=6$,

`N_TEST_SPLITS=2`) feeding a `RandomForestClassifier` (500 trees, `class_weight='balanced'` , `min_samples_leaf=20` , `max_features='sqrt'` , seed `20260930`), evaluated against a naive base-rate baseline on log-loss, accuracy, and AUC, with Mean Decrease Impurity (MDI, in-sample) and Mean Decrease Accuracy (MDA, out-of-sample permutation-based log-loss increase) feature importance. One parameter was deliberately adapted rather than copied unchanged: purge and embargo windows of **252 calendar days** (rather than this project's earlier meta-labeling work's 31 days), sized to this redesign's own longest condition lookback (`vol_regime` / `trend_strength` 's 252-day window) — a principled adaptation to this dataset's own lookback structure, confirmed before running, not a change made in response to a disappointing result. Any feature that is entirely missing (as `market_breadth_rank` was before the real breadth data arrived) is automatically detected and dropped from the active feature set, with a printed flag, rather than corrupting the run. Before being run on real data, the full pipeline was validated on synthetic data with a dedicated power check: a feature engineered to have a genuine, substantial relationship with the label is correctly picked out by MDA and beats the baseline; a feature with no relationship to the label does not.

5. Results

5.1 Stage 1 (MCPT) and Stage 2 (DSR) — 14-Asset Universe

Of 168 (asset, condition, group) cells, 13 show nominal $p < 0.05$ (uncorrected), and none survive the pooled DSR correction (`SR0` = 0.0434, shared across all 168 trials):

Asset	Condition	Group	Ann. Sharpe	p (MCPT)	PSR	Survives DSR (≥ 0.95)?
SPY	vol_regime	LOW	0.747	0.033	0.630	No
VNQ	vol_regime	LOW	0.670	0.033/0.067	0.465	No
SPY	trend_strength	HIGH	0.640	0.033	0.386	No
EEM	structural_break	HIGH	0.623	0.033	0.373	No
VNQ	market_breadth	HIGH	0.586	0.033/0.067	0.318	No
SPY	structural_break	HIGH	0.588	0.033	0.281	No
QQQ	market_breadth	HIGH	0.576	0.033	0.267	No
QQQ	structural_break	HIGH	0.532	0.033	0.195	No
HYG	credit_spread	HIGH	0.470	0.033	0.174	No
SPY	rate_environment	LOW	0.440	0.033	0.083	No
QQQ	credit_spread	LOW	0.369	0.033	0.053	No
QQQ	trend_strength	LOW	0.366	0.033	0.047	No
SPY	market_breadth	LOW	0.373	0.033	0.040	No

0 of 168 tests survive both stages. The best-placed candidate (SPY, vol_regime, LOW) reaches PSR = 0.630 against the 0.95 bar — not close.

5.2 Stage 1 (MCPT) and Stage 2 (DSR) — 10-Sector-ETF Universe

Of 120 cells, 20 show nominal $p < 0.05$. None survive the pooled DSR correction (SR0 = 0.0442, shared across all 120 trials); the best-placed candidate (XLP, vol_regime, HIGH) reaches PSR = 0.539.

Asset	Condition	Group	Ann. Sharpe	PSR	Survives DSR?
XLP	vol_regime	HIGH	0.721	0.539	No
XLP	trend_strength	LOW	0.679	0.453	No
XLI	vol_regime	LOW	0.663	0.419	No
XLF	trend_strength	HIGH	0.588	0.295	No
XLP	structural_break	HIGH	0.592	0.283	No
XLP	rate_environment	LOW	0.576	0.253	No
XLY	market_breadth	HIGH	0.562	0.235	No
XLI	market_breadth	HIGH	0.545	0.206	No
XLP	market_breadth	LOW	0.521	0.174	No
XLF	rate_environment	HIGH	0.500	0.167	No
XLF	structural_break	LOW	0.496	0.143	No

Asset	Condition	Group	Ann. Sharpe	PSR	Survives DSR?
XLI	trend_strength	HIGH	0.469	0.120	No
XLI	credit_spread	HIGH	0.436	0.087	No
XLB	structural_break	HIGH	0.423	0.075	No
XLI	structural_break	HIGH	0.416	0.071	No
XLB	market_breadth	HIGH	0.406	0.066	No
XLP	structural_break	LOW	0.378	0.048	No
XLY	trend_strength	HIGH	0.377	0.051	No
XLB	rate_environment	LOW	0.360	0.044	No
XLE	market_breadth	LOW	0.302	0.020	No

0 of 120 tests survive both stages — independently confirming the 14-asset universe's own null on a structurally different universe.

5.3 Stage 3 — Meta-Labeling

Universe	Trades used	CV paths	Beats baseline (log-loss)	Beats baseline (accuracy)	Mean AUC	Best mean MDA (feature)	market_breadth_rank MDA
14-asset	1,051 of 1,309	15	0/15 folds	0/15 folds	0.5472	+0.0127 (market_breadth_rank)	+0.0127
10-sector	727 of 1,051	15	0/15 folds	2/15 folds	0.5310	+0.0030 (trend_strength_rank)	-0.0010

For reference, before real market-breadth data was available (5 active features, `market_breadth_rank` all-missing and dropped), both universes' mean AUC sat almost exactly at chance: 0.5077 (14-asset) and 0.4985 (10-sector). Adding the real sixth feature lifted mean AUC in the 14-asset universe but left the classifier still losing to baseline in every fold there, and barely moved the 10-sector universe's own result.

6. Discussion

Three independent methods, two structurally different universes, one answer. A univariate permutation test, a pooled multiple-testing correction, and a joint nonlinear classifier are three different statistical lenses on the same underlying question, and a cross-asset universe (equities, bonds, commodities, real assets) and a single-sector-rotation universe are about as structurally different as two 10–14-name equity-market universes can be. All three lenses, on both universes, with real data throughout, land at the same answer: no conditionally exploitable edge in trend-filtered mean-reversion survives rigorous testing.

The market-breadth wrinkle, reported honestly rather than oversold. In the 14-asset universe, `market_breadth_rank` shows the single largest MDA importance seen anywhere across this entire study — an order of magnitude above the best MDA in the earlier five-feature run — alongside a mean AUC modestly above chance. Taken alone, in one universe, this could look like the beginning of something. Two facts argue against reading it that way. First, the classifier still loses to the naive baseline in every one of the 15 folds in that same universe — a positive feature-importance signal inside a model that still fails its own pass/fail bar is not the same as a working model. Second, the same feature's importance is actually *negative* in the 10-sector universe, where the classifier also loses to baseline in every fold. A feature that helps in one universe and hurts in the other, in a model that fails its actual test in both, is the signature of noise finding one lucky draw across a six-feature, two-universe search — not a replicated finding. It is reported here as a descriptive footnote, consistent with this project's practice of flagging every deviation rather than only the convenient ones, and — per the standing directive — it is not being used as a reason to revisit the model, the cross-validation scheme, or the feature design.

What a null result does and does not establish. This gauntlet does not show that markets are unconditionally efficient, nor that trend-filtered mean-reversion can never be useful under any circumstances. It shows something narrower and more defensible: within this specific pre-registered design — this rule, these six conditioning variables, this feature representation, this model class, on these two universes — no conditionally exploitable structure was detected that survives correction for the number of hypotheses actually tested. A different rule, a different set of conditioning variables, or a fundamentally different asset class could behave differently; this study does not speak to those cases.

The value of the discipline itself. The realistic alternative to a pre-registered, fixed-method gauntlet is iterative data-snooping: try a feature, see a disappointing result, try a different feature or threshold, repeat until something clears $p < 0.05$, and report only the winning attempt. That process is exactly what the Deflated Sharpe Ratio exists to catch, and it produces exactly the kind of false discovery this project has already found and rejected at earlier stages (Section 1). Holding the method fixed and reporting a clean, fully pre-registered null — even a third one, even after being specifically asked to look harder via meta-labeling — is itself the more valuable research output, not a consolation prize for having failed to find something.

A note on how conservative this null is. The DSR correction's extreme-value benchmark formally assumes approximately independent trials; because many of the pooled trials here are not independent (Section 7), the benchmark likely overstates the true bar, making the test harder to pass than a correlation-adjusted version would be. The honest characterization of this report's verdict is therefore **a null under a conservative DSR, not a null under a lenient standard** — if anything, this cuts in favor of trusting the null rather

than against it, since it would only matter if the result had been a narrow miss, which it was not (the best candidates sit at PSR 0.630 and 0.539 in the two universes, far from the 0.95 threshold).

7. Limitations

- **Coarse permutation-test resolution.** `NREPS=30` means every p-value sits at a multiple of $1/30 \approx 0.033$ — considerably coarser than the project's standing 2,000-trial convention, adopted for compute-cost reasons specific to the `structural_break` condition's SADF evaluation. Cross-checking against two informally-run alternative trial counts (34, 37) before settling on this fixed convention showed a stable "core" of recurring significant cells and a flickering edge, suggesting the coarse resolution does not change the overall qualitative picture, but individual borderline p-values in Section 5 should not be read as more precise than they are. Since SADF is the only reason the full six-condition surface cannot run at 2,000 trials, a targeted follow-up — rerunning the other five conditions alone at `NREPS=2000` — is a low-cost robustness check that has not yet been performed.
- **Trial correlation within each DSR pool, and its likely direction.** HIGH and LOW groups within one condition are complementary partitions of the same asset's own trades, and the same asset's trades recur across all six conditions, so the 168/120 pooled trials are not fully independent draws. The DSR's extreme-value-theory benchmark (`SR0`) is derived under an assumption of independent trials; when trials are positively correlated, the true expected maximum Sharpe ratio across them is lower than the formula's nominal-N calculation implies, meaning `SR0` as computed here likely overstates the true bar — biasing the test toward being too conservative (more likely to produce a false negative than a false positive). The null reported in Section 5 is therefore more precisely a null **under a conservative DSR**, not a null under a lenient standard. Given how far even the best candidates fall short of 0.95 (PSR 0.630 and 0.539 in the two universes), a correlation-adjusted, smaller effective trial count is very unlikely to reverse the verdict, but the direction of this bias should not be left ambiguous.
- **Two universes, not an exhaustive test of external validity.** A cross-asset ETF panel and a sector-ETF panel are structurally different, but both are diversified, liquid, long-only-tradeable US-listed ETF universes. A genuinely different asset class (single-name equities, FX, digital assets) was not tested and could in principle behave differently.
- **A single, fixed purge/embargo window.** The 252-day purge/embargo choice in Stage 3 is a principled adaptation to this dataset's longest condition lookback, but it was not swept across alternative windows. A longer embargo also shrinks the effective training sample available to each fold (1,309/1,051 trades is not large to begin with), so part of the classifier's loss to baseline could in principle reflect a smaller effective training set

rather than a genuine absence of joint structure — a second low-cost robustness check (rerunning Stage 3 at, e.g., 63- or 126-day purge/embargo) has not yet been performed. This would not overturn the overall conclusion on its own, since Stage 1's MCPT is independently null, but it is a natural next check.

- **One feature representation among several reasonable choices.** Expanding percentile rank was chosen over raw absolute values and over the binary HIGH/LOW label for stated reasons (Section 4), but it is not the only defensible representation; a different transform was not tested.
- **No single combined correction across all three stages.** Each stage's pass/fail bar (MCPT $p < 0.05$, DSR PSR ≥ 0.95 , meta-labeling beats baseline) is evaluated on its own pre-registered terms; running three lenses on the same underlying question is not itself formally treated as one larger family-wise test.
- **No transaction costs.** All results above are gross of transaction costs, consistent with this project's practice throughout. Since the gross result is already a decisive null, a net-of-cost result cannot look better — costs would only push an already-losing comparison further negative. This limitation is noted for completeness and cannot overturn the conclusion in the optimistic direction.

8. Conclusion

Every stage of the pre-registered gauntlet was run to completion, on both universes, with real data throughout: 168 and 120 permutation tests, two pooled Deflated Sharpe Ratio corrections, and two purged-cross-validated meta-labeling runs. The verdict is unanimous and decisive: **no conditional edge in trend-filtered mean-reversion, across six market-state conditions, survives rigorous statistical testing on either a 14-asset cross-asset universe or a 10-sector-ETF universe.** The one feature that showed an interesting signal in isolation — market breadth's feature importance in the 14-asset universe — does not replicate in the second universe and does not change either universe's own pass/fail outcome.

This conclusion is scoped specifically to liquid, long-only, heavily-arbitraged, US-listed ETF universes; it is not a claim about market efficiency in general, nor does it extend to asset classes with different liquidity, participation, or arbitrage dynamics — single-name equities, FX, or digital assets — which this study did not test (Section 7).

As a general lesson beyond this specific rule: finding a genuine, exploitable, statistically robust alpha is hard, and it should be. A rule that clears a permutation test alone, or a feature that shows promise in one universe alone, is exactly the kind of result a less disciplined process would report as a discovery. Holding the pre-registered method fixed through three consecutive nulls — rather than treating each null as a cue to try something else — is what makes this report's conclusion trustworthy rather than merely convenient.

References

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Appendix: Pre-Registered Design Decisions and Flagged Deviations

In keeping with this project's practice of flagging every deviation from an established default rather than making it silently, the table below records every point where this redesign departed from either a broader convention or the project's own prior practice, and why:

Decision	Established default	This study	Rationale
MCPT trial count	2,000 (this project's standing convention)	30, fixed constant, seed <code>20260930</code>	<code>structural_break</code> 's SADF evaluation cost makes 2,000 trials infeasible within budget; a live-benchmarked trial count was tried first and rejected after it produced machine-dependent, non-reproducible results (Section 4)
Meta-labeling purge/embargo	31 calendar days (this project's own earlier meta-labeling work)	252 calendar days	Sized to this redesign's own longest condition lookback (<code>vol_regime</code> / <code>trend_strength</code>), not the earlier work's monthly-rebalance cadence

Decision	Established default	This study	Rationale
Meta-labeling feature representation	Binary HIGH/LOW label (consistent with Stages 1-2)	Expanding percentile rank	Preserves magnitude/distance-from-median information a binary split discards, while keeping the per-asset-relative normalization the median-split design was built around
Response to null results	—	No strategy or method change made at any stage	Standing project directive: results are reported as obtained, not used to trigger a search for a different feature, model, or strategy

Data-quality handling, `market_breadth` construction. The CRSP constituent price pull required explicit, non-default handling of three data patterns not previously encountered on this project's smaller, hand-picked 14-asset universe: a `NaN dlyprc` (no quote that day) is left as a missing observation rather than interpolated or dropped as a row; a negative `dlyprc` (CRSP's own bid-ask-average, no-trade convention) is recovered via absolute value rather than treated as an error; a missing or non-positive `dlycumfacpr` (split-adjustment factor) produces a missing, rather than a divided-by-an-invalid-number, adjusted price. In every case the missing observation flows through as "not eligible that day" in the downstream breadth calculation, consistent with this project's point-in-time-membership discipline, rather than being silently filled.